

Pre-Computing Pattern

 systemdesignschool.io/fundamentals/pre-computing-pattern



Grasping the building blocks ("the lego pieces")

This part of the guide will focus on the various components that are often used to construct a system (the building blocks), and the design templates that provide a framework for structuring these blocks.

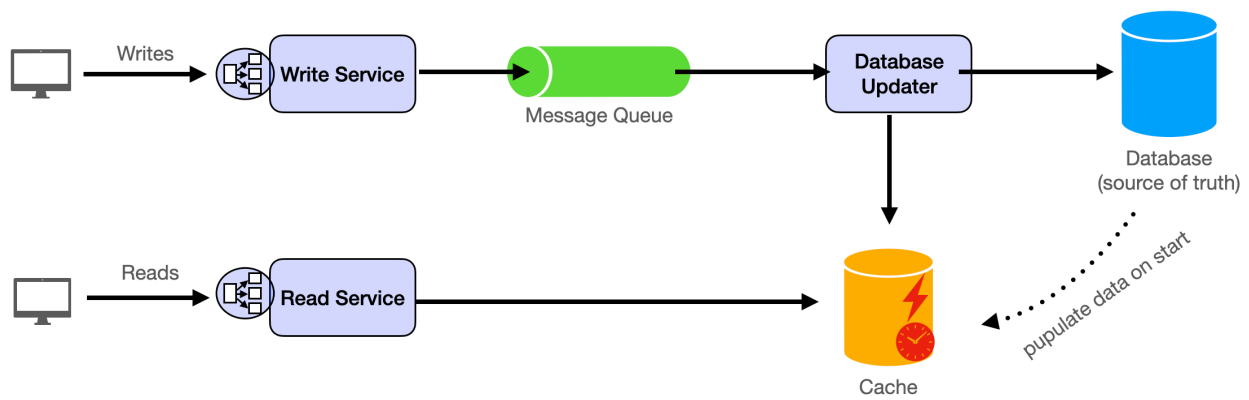
Core Building blocks

At the bare minimum you should know the core building blocks of system design

- Scaling stateless services with load balancing
- Scaling database reads with replication and caching
- Scaling database writes with partition (aka sharding)
- Scaling data flow with message queues

System Design Template

With these building blocks, you will be able to apply our template to solve many system design problems. We will dive into the details in the [Design Template](#) section. Here's a sneak peak:



Additional Building Blocks

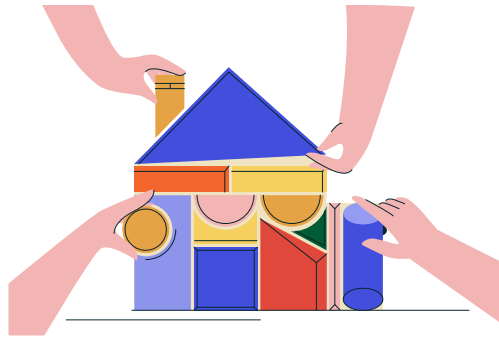
Additionally, you will want to understand these concepts

- Processing large amount of data (aka “big data”) with batch and stream processing
Particularly useful for solving data-intensive problems such as designing an analytics app
- Achieving consistency across services using distribution transaction or event sourcing
Particularly useful for solving problems that require strict transactions such as designing financial apps
- Full text search: full-text index
- Storing data for the long term: data warehousing

On top of these, there are ad hoc knowledge you would want to know tailored to certain problems. For example, geohashing for designing location-based services like Yelp or Uber, operational transform to solve problems like designing Google Doc. You can learn these on a case-by-case basis. System design interviews are supposed to test your general design skills and not specific knowledge.

Working through problems and building solutions using the building blocks

Finally, we have a series of practical problems for you to work through. You can find the problem in /problems. This hands-on practice will not only help you apply the principles learned but will also enhance your understanding of how to use the building blocks to construct effective solutions. The list of questions grow. We are actively adding more questions to the list.



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